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(1) Research Program/Project/Study Title: Enhancement and Pilot Testing of a Multi-Layer Fish Dryer with Biomass Furnace	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Agri-Processing/Fish	(8) Discipline/Field of Specialization Agricultural Engineering
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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Sun-drying is the widely used method for drying fish, however, this method results in a relatively slow drying rate, thus the tendency of spoilage. On the other hand, dryers that use LPG and electricity have emerged. However, these dryers are associated with higher operating costs. Hence, this study aims to develop a multi-layer fish dryer that uses rice husks as the heat source. The dryer comprises a biomass furnace with a heat exchanger, drying chamber, display, and control panel. The initial design was subjected to computational fluid dynamics simulation before fabrication to ensure uniformity of drying temperature in the chamber. For drying temperature adjustment, the control panel comprises a combustion controller with selections such as low (41-50°C), medium (51-60°C), and high (61-70°C). The performance of the fabricated dryer was tested in terms of drying time, drying rate, heating system efficiency, and moisture content of the dried product. Results showed that in drying *Sardinella longiceps* using high temperatures, the drying time is 7.5 hours, the drying rate is 8.62 kg/hr, and the heating system efficiency is 64.29%. The final moisture content of fish is 35.51%. The performance data gathered during the pilot testing of the dryer shows that the required moisture content varies depending on the type of dried fish. The final moisture content ranged from 20 to 45%. Hence, the drying time, fuel consumption, and dried weight recovery for each type of fish also varies. Due to the difference in the sizes and surface area of the fish tested, the maximum input load per drying batch also varied from 52 to 124 kilograms per batch. The test conducted by the Agricultural Machinery Testing and Evaluation Center showed that the heating system efficiency of the dryer passed the standard requirement based on the Philippine Agricultural Engineering Standards, hence, the developed dryer offers an efficient technology for fish drying. Likewise, transferring the technology to its users may significantly improve the production of dried fish in terms of food safety.

Keywords: *fish dryer, rice husks, drying, biomass furnace*

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(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Fish is a highly perishable food that becomes rotten within a few hours after capture. Hence, the need for preservation and processing to reduce microbial decomposition and retard spoilage. "In the Philippines, fish drying is the most practiced processing method, especially for the small-sized species" [1]. It was found that reducing the moisture content of fish to 25% will retard bacteria and reduce autolytic activity.

Sun drying is the widely used method for fish drying as it requires cheaper operating expenses. However, this method has disadvantages due to unpredictable weather conditions, resulting in uneven drying and a relatively slow drying rate, thus the tendency of spoilage. Hence, several dryers have been developed to dry fish products. Dryers that use a combination of solar and biomass as a source of energy have emerged. Also, the "use of LPG/electricity has substantially increased in the last few years" for this purpose [2]. However, these drying heat sources are associated with higher initial and operating costs. Thus, they have not been used in the industry considerably.

Biomass is an effective alternative to traditional sun-drying and dryers, which use expensive electricity and fossil fuels for their heat source. It is a clean and environmentally friendly fuel and does not cause any pollution during combustion. It has been used successfully as a fuel in many agricultural machines – including a flatbed dryer for grains. It entails cheaper investment and operating costs. Rice husk, a by-product of rice milling, is an attractive biomass fuel because it is readily available in large quantities and easy to collect.

The prior arts on fish drying systems involve different designs and encompass a common principle: the "hot air absorbs and conveys moisture from the product by heat and mass transfer and diffusion while the residual air is discharged through an exhaust unit." The most common types are cabinet dryers consisting of several layers of fish trays. Non-uniformity of dried products in terms of final moisture content is a common problem with this type of dryer. The flatbed dryer is another type and was introduced for higher drying efficiencies. Ideally, however, flatbed dryers are designed to dry materials with smaller surface areas, are porous, and have an initial moisture content (MC) of 25 to 30%, such as grains. Case hardening is a disadvantage for fish with higher MC (up to 80%) and larger surface areas. Due to higher space requirements, the capacity (load/batch) would be another limitation in this type of dryer. Therefore, a study needs to be conducted to modify the existing dryers.

Hence, CAPSU researchers designed and fabricated a flatbed-type biomass-fired fish dryer with twenty kilograms (20 kg) capacity from 2020 to 2021. It was evaluated using *Manamsi* (*Sardinella albella*). Results showed that even distribution of hot air in terms of temperature, airflow, and velocity in the drying bed was achieved with the present design. It could dry fish samples to 25 to 30% MC (initial MC = 60 to 70%). Higher drying rates were also achieved than in the previous studies/designs at only about eight (8) hours of operating time per batch. In addition, the products of biomass combustion did not come in contact with the fish being dried, thus did not affect the quality of the dried products.

In an initial technology presentation with the fish processors in the Province of Capiz, they commended the fish dryer project of CAPSU. However, according to them, the 20-kg capacity

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per batch should be tripled or even increased to 100 kilograms to satisfy the volume requirement of small to medium-scale processing. They added that varied fish types/species are also processed and should be tested. Also, the system efficiency of the equipment was 49%, which is lower than the Philippine Agricultural Engineering Standards (PAES) requirement, which is 50% for indirect-fired dryers.

To increase the capacity of the prototype (1), the researchers improved its design into a bed-cabinet-type dryer. Given the constraints of cabinet-type or layer-type dryers, the researchers initially designed and fabricated a forty-kilogram (40-kg) fish dryer with minor modifications to the prototype (1). The essential mechanisms/components of the prototype (1) were applied in the modified prototype (2).

With the study results and to provide dried fish processors with an advantageous drying technology for drying fish, the researchers developed and pilot-tested a multi-layer fish dryer using rice husks as a heat source.

Specifically, the study aims to modify and improve the previous prototype in terms of capacity (up to 100 kg) and system efficiency (>50%), evaluate the performance in terms of drying air temperature uniformity, final moisture content of dried products, drying time, drying rate, and heating systems efficiency, test its performance through the Agricultural Machinery Testing and Evaluation Center (AMTEC), and conduct a pilot testing.

This project is expected to help increase the production capacity and product quality of local dried fish processing and increase the utilization of biomass wastes. It will directly benefit the dried fish processors-cum-traders in the province and its neighboring province by providing them with a dryer technology that is hygienic and efficient. Likewise, local machine fabricators may benefit from the project during the machine's commercialization. Indirectly but essentially, it will be helpful to fisherfolks by ensuring fish catch demand even during the rainy season and to the consumers as it may significantly stabilize dried fish prices.

Lastly, this project will improve scientific dissemination and serve as a direct source of knowledge or ideas related to cabinet-bed-type dryers that use biomass for more advanced technological possibilities.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

A. Design and Fabrication of the Dryer

a. Design Considerations

The dryer's target capacity is one hundred (100) kilograms or three (3) *banyeras* per batch. The volume of the drying cabinet has been calculated considering the size of *Manamsi*, specifically its surface area.

The dryer's biomass furnace is designed to meet target drying time of eight (8) hours per batch, considering the initial moisture content of *Manamsi* at eighty percent (80%) and the target final moisture content of dried products at thirty five to forty percent (35-

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40%). The recommended maximum drying air temperature of up to seventy degrees Celsius (70°C) is considered in calculating the drying heat requirement.

The dryer is designed to provide insulation and uniform air to meet the required heating system efficiency of more than fifty percent (>50%). The dryer's insulation size is calculated considering the properties of selected insulation materials. To ensure uniform drying air, the initial design is subjected to simulation, specifically the Computational Fluid Dynamics. Design iterations are made until simulation provides better results in terms of drying air temperature uniformity.

b. Dryer Fabrication

The multi-layer fish dryer comprises the following major components: the biomass furnace, expanding air duct, control and indicator, drying cabinet, exhaust vents, and backdraft damper (Figure 1).

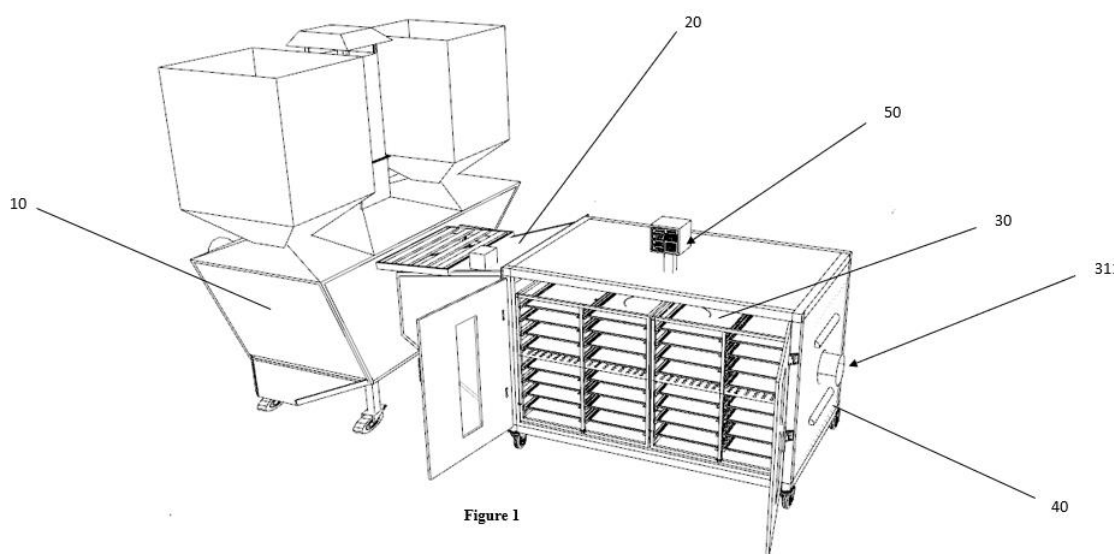


Figure 1

Legend:

- 10 - biomass furnace
- 20 - expanding air duct
- 30 - drying cabinet
- 31 - backdraft damper
- 40 - exhaust vents
- 50 - control and indicator

Biomass Furnace. The biomass furnace serves as the dryer's heat source. It is being held by a support structure with a horizontal and vertical member. The biomass furnace provides two biomass holding bins that store biomass material for continued operations. The biomass holding bins were disposed of in the furnace's topmost portion. Each holding bin has a total volume of 12 to 18 cubic meters per bin built into four-sided bins. The biomass holding bins

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have a hopper, defined by an inverted pyramid or a spout directly below each bin. It serves as a rice hull outlet from the biomass holding bins and goes into the combustion area. The holding bins and the hopper have a partition sheet in between them, which is being opened to feed rice hulls into the combustion area during the operations. A feeding mechanism is being installed adjacent to the hopper of the holding bins. The feeding mechanism is defined by an inclined plate positioned at a 42-degree angle toward the combustion area. This design allows the feeding biomass fuel into the combustion area by gravity. The holding bins simultaneously provide biomass to the combustion area through the said feeding mechanisms from above the two sides of the said combustion area. A combustion area is found at the center and lower portion of the biomass furnace. The said area receives biomass fuels for combustion. The walls of the combustion area have been insulated with clay bricks to minimize heat loss. A grate holds the biomass fuels and releases ash every 15 minutes during combustion. The said grate has a lever configured to move in half rotation motion, thereby discharging ash into the holes of the grate. To allow the starting up of biomass fuel, a combustion area door is removably fixed outside the combustion chamber. A combustion controller is installed on one side of the combustion area to assist in biomass combustion. It is installed through an air channel to ensure pressure for uniform air distribution for efficient burning. The combustion controller has an attached speed regulator. The speed regulator provides varied air flow rates. One important component of the combustion area is the heat exchanger set. The heat exchanger set is disposed at the upper portion of the combustion area. The opening ends of the hollow cylinders are exposed at the sidewalls of the walls of the combustion area. An air vessel conceals together the opening ends of the hollow cylinders on one of the sidewalls of the combustion area. A blower is fixed to the end of the said air vessel to provide clean air into the hollow cylinders. The blower has a diameter of 25 centimeters, running at 2,800 rpm and providing 1,100 cubic meters of air per minute. The blower has been adapted to provide sufficient airflow rates as a drying medium for fish.

Air duct. An expanding air duct is attached to the hollow cylinders' other opening sides to receive clean air blown by the fan to the hollow cylinders and heated by the combustion. The expanding air duct width expands until it accommodates the size of the drying cabinet. The expanding air duct is attached to one opening side of a drying cabinet. An essential component of the air duct is a baffle set installed in the mid portion of the air duct, resulting in an equal air distribution inside the drying chamber. The baffle set is composed of 9 air grilles being moved by a motor through a lever arm to move the 9 air grilles in reciprocating motion from a 70-degree angle up to a 90-degree angle down for twenty seconds.

Drying cabinet. The drying cabinet is supported by a support frame defined by a plurality of vertical and horizontal framing members to carry the drying cabinet securely. The drying cabinet comprises an insulated enclosure and an insulated door to minimize heat loss inside the drying cabinet. Fiber mats or coconut fibers having a thickness of 0.75 inches may be used as insulation material. The insulation was preferred to increase drying efficiency by ensuring minimum heat loss during drying operations. The insulation thickness has been calculated as follows:

Formula:

$$Q = \lambda \Delta T / L A$$

Where:

Q = Total heat

λ = thermal conductivity of insulation material (fiber mat)

L = thickness of insulation

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A = Total wall area

Therefore:

$$L = \lambda \Delta T A / Q$$

$$L = ((0.14 \text{ W/mK})(85-25) (11.64 \text{ m}^2)) / (3,000 \text{ W})$$

$$L = 0.0326 \text{ m} = 1.28 \text{ in} \approx 1.5 \text{ in}$$

The drying chamber of the initial design has been tested and enhanced using Computational Fluid Dynamics (CFD) Simulation to ensure uniformity of air in the drying chamber, thereby enhancing Heating System Efficiency and dried product quality. Several design iterations/adjustments were made to ensure equal air distribution regarding air temperature and air velocity. These adjustments include re-sizing the chamber, adding air baffles, automating the burner fan, and re-designing the exhaust. Figure 2 presents the simulation of the initial design (a) and the simulation of the final design (b). It can be observed that the drying air is not evenly distributed throughout the chamber. Thus, the dryer has been re-designed considering a more efficient exhaust, longer air duct, addition or provision of baffles at the air duct, lessen the dryer length by 20%, and increase the height of the dryer or the number of layers. A better air distribution is observed in the simulation results of the final design.

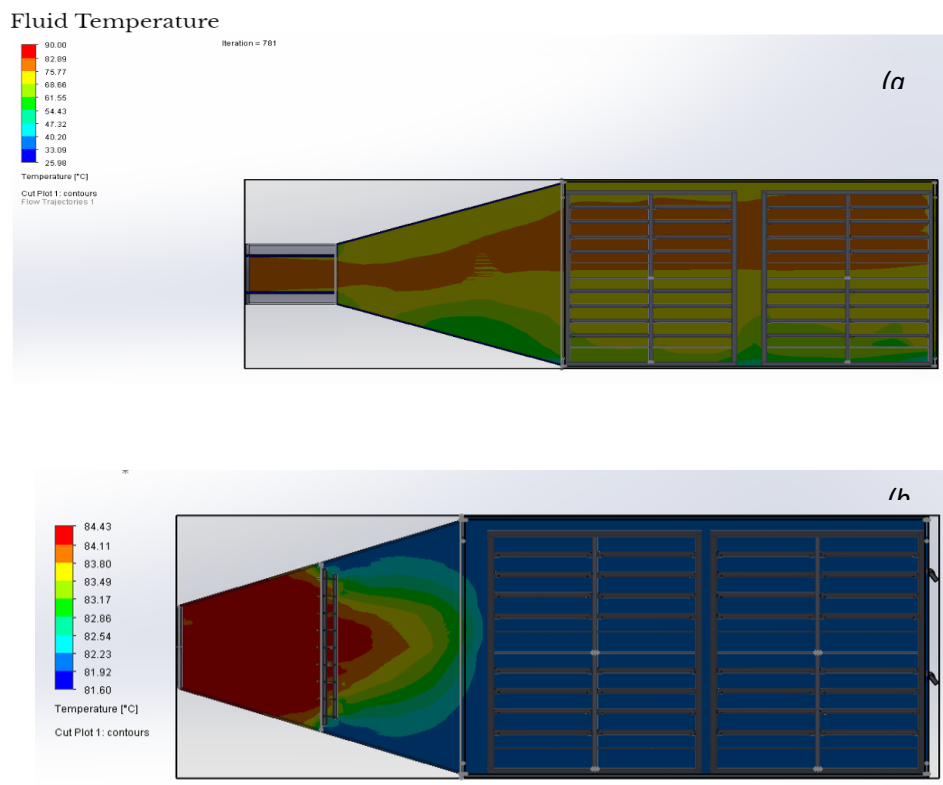


Figure 2. Temperature contours of the initial design (a) and the final design (b).

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The drying cabinet consists of drying tray racks and drying trays. The drying tray racks hold the drying trays at a vertical distance of 2 to 3 inches from each other to ensure that each removable drying tray receives drying air during the drying operations. It has a deflector sheet disposed of at the centermost and lowermost part of the drying cabinet. The deflector sheet allows uniform distribution and velocity of drying air into the drying trays, especially in most side portions of each tray. The deflector sheet has a plurality of air holes. The dryer also consists of exhaust vents. The exhaust vents have a width of 1 inch and a length of 32 inches and are formed at the leftmost portion of the shorter side of the insulated enclosure.

Backdraft damper. A backdraft damper has been installed between the two exhaust vents to ensure moist air does not accumulate in the drying cabinet. The said backdraft damper drives moist air out of the drying cabinet efficiently, which aids in efficient product drying. The damper door opens to release moist air efficiently.

Drying control system. A drying control system has been mounted on the drying cabinet. Said drying control system has a provision of external temperature and RH sensors placed in two areas of the drying cabinet; the external temperature and RH sensors are linked to an indicator on said drying control system, configured to display parameters of temperature and RH to allow readings or external display; the drying control system further has:

- a start control button connected to the blower;
- a temperature process control input button was adapted to allow users to input the desired drying temperature, such as low, medium, and high, and
- an RH process control input button adapted to input desired RH.

**c. Performance evaluation of the fabricated
Multi-layer fish dryer**

After the fabrication, the dryer underwent a running-in period for 24 hours, where important parameters such as drying air flow rate and velocity and drying temperatures (in the different points of the drying cabinet were observed and recorded. Various calibrations and adjustments were made to ensure the machine's performance. The rice hull burning rate and the drying temperatures were checked and calibrated before the performance evaluation trials. Several thermometers were properly installed in the different parts of the dryer and the ambient air for recording. An airflow meter was also installed at the airflow duct.

The drying temperature plays a crucial role in maintaining the final product's appearance. Hence, different drying temperatures have been tested to effectively mitigate the defects as described in PNS/BFAD 04:2006. The dryer was tested using *Sardinella fimbriata* (Tunsoy) fish at 70, 65, 60, and 55 degrees Celsius. The product outputs from the said tests were observed in terms of appearance.

The dryer was tested using 55 degrees Celsius drying air temperature for three (3) trials using fresh *Sardinella fimbriata* (Tunsoy) for each trial. The duration of each trial started after pre-heating and ended when the fish's desired moisture content (35 to 40%, based on Fish Processing guidelines of the DOST-ITDI) was attained.

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Before the start of each trial, a hundred (100) grams of fish samples were collected to determine the initial moisture content of the fish by oven-drying method with a 103°C drying temperature for twenty-four (24) hours. Two hundred fifty (250) grams of representative samples from each of the four trays per layer were marked and weighed every 30 minutes to determine weight loss. The drying process ended when the desired weight loss was achieved in the representative samples. After each trial, the dried fish were collected and weighed to determine the percent loss in moisture content.

In the performance evaluation, the following data and parameters will be gathered and calculated based on the Philippine Agricultural Engineering Standards – Methods of Test in Mechanical Dryers:

- drying air temperature – gathered at the different points of the drying cabinet and monitored at intervals every hour.
- the initial and final weight of the samples - refers to the weight of the fish samples before the drying process, while the final weight refers to the weight of the fish after the drying process.
- Changes in the samples' weight after every thirty-minute interval are determined by subtracting the final weight from the initial weight.
- drying rate - refers to the amount of moisture (kg/hr) removed from the fish per unit of drying time.
- initial and final moisture contents – refer to the amount of moisture in the fish/dried fish, calculated by subtracting the dry weight from the wet weight, dividing the result by the wet weight, then multiplying by 100.
- heating system efficiency (HSE_#) - refers to the ratio (%) between the heat output and the heat input in the system. It was obtained using the formula:

$$HSE_{\text{Eff}} = \frac{(\text{Enthalpy 1} - \text{Enthalpy 2})(\text{Air Flow Rate} \times \text{SP. Volume})(\text{Drying Time})}{(\text{Fuel Consumed} \times \text{Heat Content of Fuel})}$$

- The airflow rate was determined by measuring air velocity using an anemometer multiplied by the duct's cross-sectional area.

d. AMTEC Testing

The Agricultural Machinery Testing and Evaluation Center (AMTEC) tested the dryer based on PNS/BAFS 344:2022 Agricultural Machinery- Agricultural and Fishery Commodity Dryer – Methods of Test. The dryer was tested last February 5, 2024, at the Capiz State University, Brgy. Burias, Mambusao, Capiz.

The different parameters tested were drying capacity, moisture reduction rate, Heating System Efficiency, heat utilization, Drying Efficiency, Drying System efficiency, Moisture Content Variation, Moisture Content Gradient, Fuel Consumption/Power Consumption, Number of Operators, and Quality of Dried Material.

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e. Pilot Testing and Feasibility of using the dryer

Pilot testing of the developed dryer has been conducted in Dumolog, Roxas City, Capiz, through the Dumolog Dried Fish Association with approval from its Chairperson, Mr. Ricky Bacolod (Appendix 5). The focal person in the said pilot testing was Mrs. Yolly Escanio, and the assigned operator was Mr. Ronel Bosico. The dryer was stationed in the vacant lot of Mr. Mario Agase, who is also one of the large-scale dried fish operators in the area.

A project presentation was conducted by the project team members to the City Government of Roxas City with Mayor Ronnie T. Dadivas. The presentation is hoped to result in a partnership with the said City Government to commercialize the developed biomass-fired fish dryer (Appendix 6-7).

Appendix 8. Photos during the presentation of the biomass-fired fish dryer to the Dumolog Dried Fish Association members, prior to pilot testing.

The abovementioned association tested the dryer in drying different types of fish, such as:

- Medium Type Anchovies "Dilis" (*Stolephorus comersonii*)
- Large Type Anchovies "Dilis" (*Stolephorus comersonii*)
- Boneless Anchovies "Dilis" (*Stolephorus comersonii*)
- Splitted Golden caesio "Dalagang bukid" (*Caesio chrysozonus*)
- Splitted Red Mullet "Salmonete" (*Mullus surmuletus*)

During the pilot testing, the Operator's Acceptability was gathered. Operator's acceptability is the overall rating of the operator in terms of ease of using the machine. It shall be rated as 1- Very good, 2- Good, 3-Satisfactory, 4-Poor, and 5- Very poor. After operating the dryer, the three operators rated different observations (see Table 4).

Cost details were gathered, and the total production cost and gross profit were calculated.

f. Physicochemical Analysis of Dried Fish

The laboratory analysis of dried fish processed using the dryer and the sun-drying methods has been studied through the Regional Standards and Testing Laboratory (RSTL) of DOST Region VI.

Tests such as Water Activity and NaCl as Salt have been determined.

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(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

A. Performance of the dryer in terms of defects in appearance and other drying parameters.

The dryer was initially tested using *Sardinella fimbriata* (Tunsoy) fish at 70, 65, 60, and 55 degrees Celsius. The product outputs from the said tests were observed in terms of appearance, and the results are presented in Table 1.

Table 1. The presence of defects in the appearance of output at different drying air temperatures

Type of defect (PNS/BFAD 04:2006)	Temperature, °C			
	70	65	60	55
1. Loose scales	/	/	/	X
2. Presence of liver and blood stains and traces of internal organs	X	X	X	X
3. Bursting of bellies	X	X	X	
4. Excessive cracks and crumbling texture	/	X	X	X
5. Detaching of fish parts	/	/	X	X
6. Excessive salt crystals appearing on more than 50% of the fish surface	X	X	X	X

X- no presence, /- with presence

The data reveals that drying fish at temperatures of 60, 65, and 70 degrees Celsius leads to several undesirable defects in appearance. These defects include loose scales, excessive cracks and crumbling texture, and detaching of fish parts. Based on the data highlighting defects at higher temperatures, a drying temperature of 55 degrees Celsius is adopted in further testing the dryer. At this temperature, no defects are observed in the dried fish output. The drying performance of the dryer in terms of capacity, drying rate, drying time, and heating system efficiency are reflected in Table 2.

Table 2. Performance of the dryer.

Parameters	Trial			Mean
	I	II	III	
Weight of input (kg)	137.10	133.30	128.30	132.90
Weight of output (kg)	66.93	68.90	68.80	68.21
Weight of moisture removed (kg)	70.17	64.40	59.50	64.69
Drying Time (min)	450.0	465.0	435.0	450.00
Drying Rate (kg/hr)	9.4	8.3	8.2	8.62
Total weight of fuel utilized (kg)	137.8	143.4	113.1	131.43
Fuel utilization rate (kg/hr)	18.38	18.50	15.60	17.52
Heating System Efficiency	63.69%	66.10%	63.08%	64.29%

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The average capacity per batch of the dryer is 132.90 kilograms, and the average output per batch at the target moisture content (35 to 40 degrees Celsius) is 68.21 kilograms. Therefore, the water removed per batch is 64.69 kilograms. The drying time per batch is 450.00 minutes (or 7.5 hours); hence, the drying rate is 8.62 kilograms per hour.

The fuel consumption rate is 17.52 kilograms per hour.

The average Heating System Efficiency is 64.29%, considering the heat supplied by the burner, the fuel consumption, and the heat content of the fuels. This value is considered higher than the standard for indirect-fired dryers, which is 50% (Philippine Agricultural Engineering Standards) and the previous prototype. This indicates an improvement in how effectively the dryer converts fuel into heat energy for the drying process.

This was higher than the required heating system efficiency for indirect-fired dryers, which is 50% based on Philippine Agricultural Engineering Standards. This was also higher than CAPSU's 20-kilogram fish dryer, with only 49% efficiency. This can be attributed to the design enhancements in the drying and combustion chambers.

The average drying air temperature in the various points inside the drying chamber, gathered at an interval every hour, is presented in Figure 3. The figure shows that the air temperature ranges from 55.07 to 57.14 degrees C. The results performed better than the previous biomass-fired fish dryer of CAPSU, where drying air temperature across different drying trays ranged from 72.2 °C to 87.2 °C. The improvement is attributed to the enhanced air duct and drying chamber designs.

Figure 4 presents that drying air temperature ranges from 52.75 to 58.13°C when plotted against drying time. Likewise, it performs better than the previous dryer with air temperatures ranging from 69°C to 87°C. This is attributed to the improved combustion chamber design, which provides continued and uniform combustion.

The curve in Figure 5 indicates that the fish moisture contents decreased linearly with the drying time. The average initial moisture content is 66.87%, while the final moisture content is (35.51%).

Figure 6 shows the final moisture content of fish after drying collected from the different drying trays. The final moisture content ranges from thirty-four to thirty-six percent (34.4-36.6%).

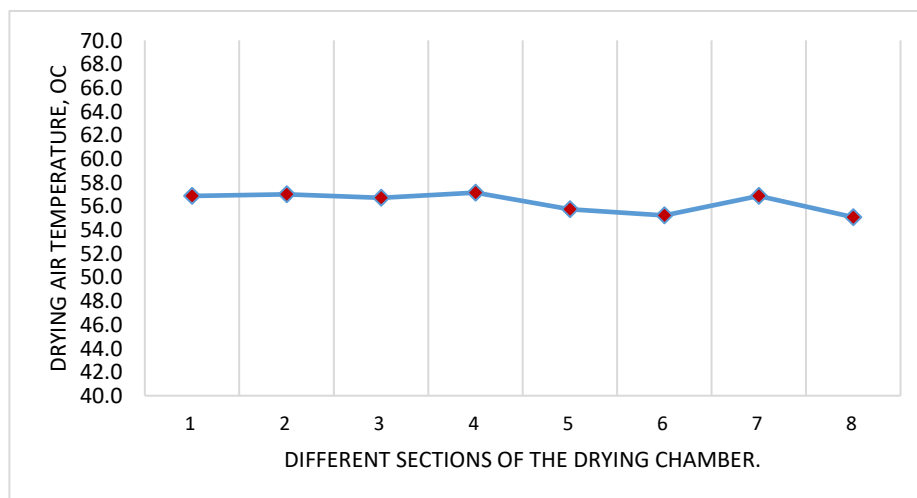


Figure 3. Average drying air temperature taken from the different sections of the drying chamber.

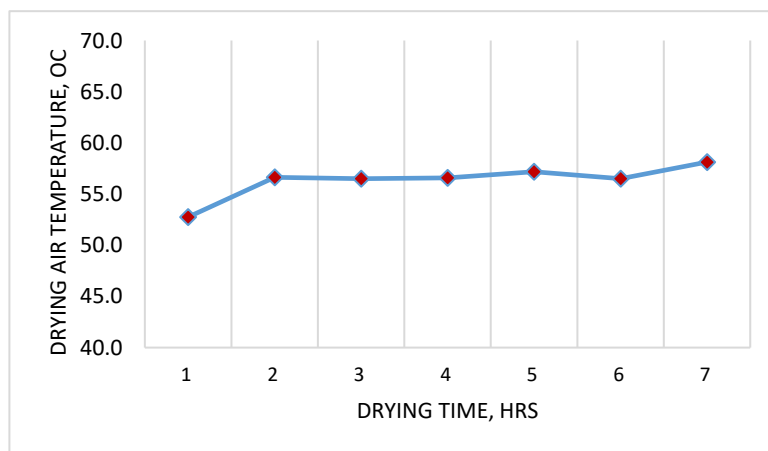


Figure 4. Average drying air temperature plotted against drying time.

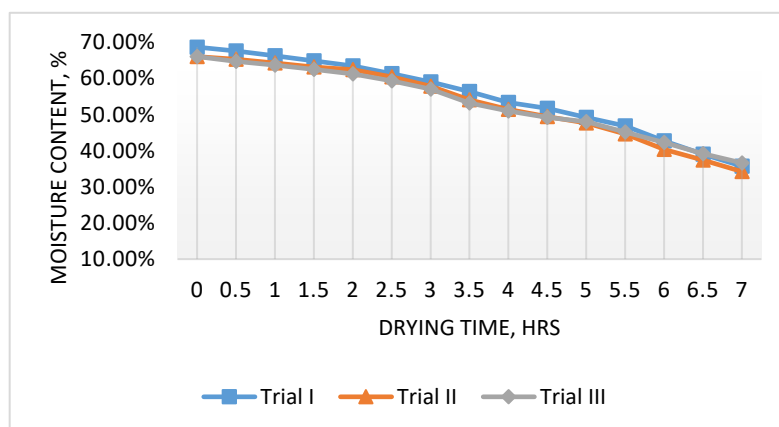


Figure 5. Fish moisture content plotted against drying time.

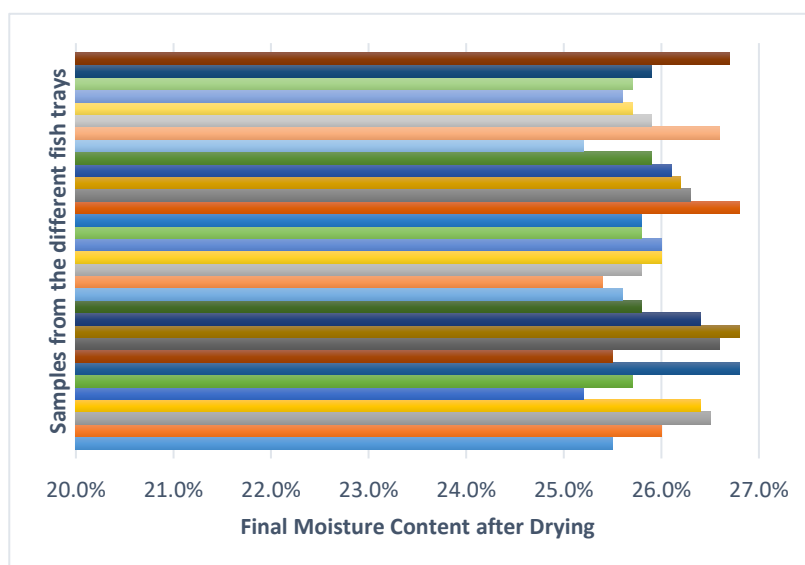


Figure 6. The final moisture content of dried fish was taken from the different drying trays.

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B. AMTEC Test Results

Figure 7 is a screenshot of the AMTEC Test Report Summary; the complete report is attached in Appendix 4. Results revealed that Heating System Efficiency is 69.36%, which is higher than the 50% required by PNS/BAFS 343:2022.

Test Report Number	2024 – P0002
Date of Issuance	March 4, 2024
Valid Until	March 4, 2029
Agricultural Machinery	Agricultural and Fishery Commodity Dryer
Type	Batch-type, Tray-type, Enclosed drying chamber, Indirect combustion, Renewable energy (Biomass energy), Tested for salted fish
Brand	CAPSU
Model	TAPICAP
Test Requested by	CAPIZ STATE UNIVERSITY Capiz State University, Brgy. Burias, Mambusao, Capiz

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 343:2022	AMTEC Test
Drying Capacity, kg/h, maximum	ND ^a	11.42
Drying Air Temperature, °C	Shall be uniform and equally distributed in the drying chamber	65.3 ± 1.63
Heating System Efficiency, %, minimum	50 ^b	69.36
Water activity, maximum	0.78 ^c	NM ^e
Moisture content wet basis, %, maximum	ND ^d	35.76

^a Not specified by the manufacturer.

^b Requirement for indirect-fired biomass fuel.

^c Requirement for dried salted fish at 25°C based on PNS/BFAD 04:2006

^d Not specified on PNS/BAFS 343:2022

^e No available test instrument to facilitate the measurement of the parameter.

Note: ND – No Data, NM – Not Measured

Figure 7. Screenshot of the Summary of AMTEC Test Report

C. Figure 8 presents the performance data gathered during the pilot testing of the dryer. The figure shows that the required moisture content varies depending on the type of dried fish and whether they are processed whole or split. The final moisture content ranged from 20 to 45%. Hence, the drying time, fuel consumption, and dried weight recovery for each type of fish also varies. Due to the difference in the sizes and surface area of the fish tested, the maximum input load per drying batch also varied from 52 to 124 kilograms per batch.

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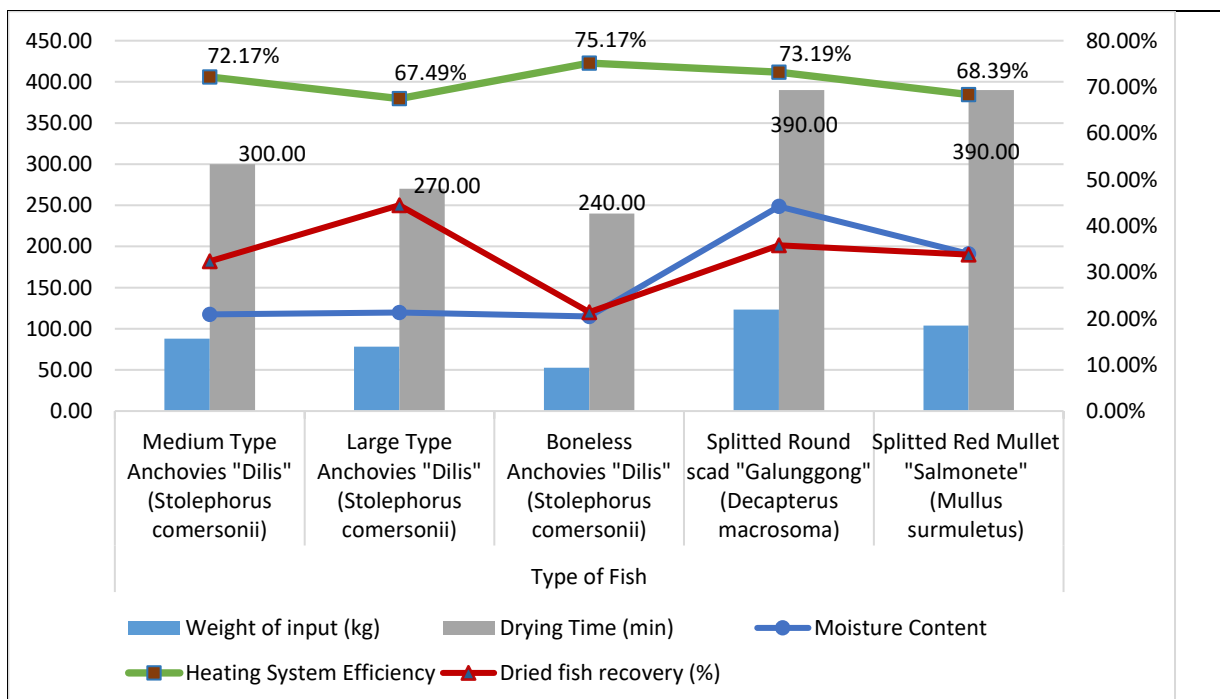


Figure 8. Performance of the dryer during the pilot testing with different fish.

Table 3 shows the dryer's air-drying condition and fuel consumption during the pilot testing. The product quality, as assessed by the processor, is also shown in the table. Results revealed that high temperatures (61 to 70 degrees Celsius) are suitable for producing good quality dried fish.

Table 3. Performance of the dryer during pilot testing.

Parameters	Type of Fish				
	Medium Type Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Large Type Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Boneless Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Splitted Round scad "Galunggong" (<i>Decapterus macrosoma</i>)	Splitted Red Mullet "Salmonete" (<i>Mullus surmuletus</i>)
Drying Air Temperature	67.00	65.00	65.00	67.00	65.00
Total weight of fuel utilized (kg)	100.00	96.25	81.25	132.50	125.00
Fuel utilization rate (kg/hr)	20.00	21.39	20.31	20.38	19.23
Product Quality	Good	Good	Good	Good	Good

During the pilot testing, the Operator's Acceptability was gathered. Operator's acceptability is the overall rating of the operator in terms of ease of using the machine. It shall be rated as 1- Very good, 2- Good, 3-Satisfactory, 4-Poor,

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and 5- Very poor. After operating the dryer, the three operators rated different observations (see Table 4).

Table 4. Operator's acceptability of the optimized pineapple leaf fiber scraping machine (P1).

Statement	Verbal Interpretation
1. Ease of loading	Satisfactory
2. Ease of cleaning the parts	Very good
3. Ease of adjusting and repair of parts	Good
4. Ease of collecting output	Very good
5. Safety	Very good
6. Vibration	Very good

Source of instrument: Philippine Agricultural Engineering Standards (PAES)

The operators were also interviewed regarding the issues and concerns and their observations of the dryer. Several concerns regarding its operation and safety were highlighted, such as:

➤ **Management of flue gases**

One of the operators explained that the main issues in operating the dryer is when the furnace gets choked with rice hull, causing excess smoke. While the chimney directs most of the flue gases away from the operators, some choking incidents can result in a noticeable increase in smoke emissions. These concerns underscore the importance of possibly exploring enhancements to prevent the furnace from getting choked. In summary, while the dryer machine is generally functional, operators' concerns regarding occasional furnace blockages leading to excessive smoke highlight areas for improvement. Addressing these concerns will enhance operational efficiency and mitigate environmental and health risks associated with increased smoke emissions.

➤ **Handling of Ash from Biomass Furnace**

During discussions with operators of the dryer, a notable concern was raised regarding handling ash after test operations. After each test operation, the operator noticed that the biomass furnace ash fell directly onto the ground. According to them, it's quite messy and needs to be manually collected using sacks. The operators were concerned about handling the ash manually, which is time-consuming and potentially hazardous due to the risk of inhaling or getting ash particles into their clothing. These concerns highlight the need for an ash collector from the biomass furnace.

➤ **Optimizing Drying Processes for Fish: Operators' Perspective**

In conversations with operators of the dryer, a key concern emerged regarding the efficiency of the drying process, specifically in relation to determining the optimal drying time for different types of fish. According to one of the operators, it is a challenge that there is no recommended drying time for each type of fish. This lack of guidance means inspecting the fish frequently to ensure they're dried properly. These insights underscore the operators' desire for more structured guidelines to streamline the drying operations for different types of fish. Establishing recommended drying times based on empirical data or

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industry standards could significantly enhance productivity and quality control in the drying process.

➤ **Positive Impacts of the Fish Dryer: Operators' Perspective**

When asked about the positive impacts of the fish dryer, operators highlighted several key aspects that have significantly improved their operations while maintaining product quality. According to the operators, an on/off switch, temperature sensors, a display showing the drying air temperature inside the drying chamber, and control buttons allow accurate monitoring and adjusting the drying conditions.

Operators also observed no undesirable odors or colors, indicating that the dryer's output is comparable with those dried through sub-drying.

The operators have highlighted considerably faster drying operations than traditional methods, which helps meet production demands, especially during rainy conditions.

The operators' feedback underscores the importance of advanced control features in enhancing efficiency and maintaining product quality in fish drying processes. The absence of undesirable odors or discoloration further validates the dryer's effectiveness in preserving the fish's natural attributes. They also highlight the dryer's ability to significantly accelerate drying times, thereby improving overall efficiency in their operations.

Production Cost and Gross Profit

During the pilot testing of the dryer, cost details were gathered, and the total production cost and gross profit were calculated. Results are presented in Table 5. The production cost and gross profit data are based on one drying batch.

Table 5. Production Cost and Profit in Using the Dryer

Parameters	Type of Fish				
	Medium Type Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Large Type Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Boneless Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Splitted Round scad "Galunggong" (<i>Decapterus macrosoma</i>)	S
Production Cost					
Cost/banyera	3,000.00	2,800.00	4,000.00	4,500.00	
Cost of fish/batch	6,000.00	5,600.00	8,000.00	13,500.00	
Salt	150.00	150.00	150.00	450.00	
Labor for laying in trays	76.80	76.80	76.80	76.80	
Labor for operator	250.00	300.00	250.00	450.00	
Cost of splitting			200.00	300.00	
Electrical Cost	22.42	24.91	19.93	32.38	
Ricehull Cost	115.50	90.00	97.50	159.00	
Total Production Cost	6,614.72	6,241.71	8,794.23	14,968.18	
Production cost/kg	190.08	219.01	779.63	338.65	
Profit					
Cost of dried fish/kg	250.00	300.00	1,000.00	450.00	
Cost of Goods Sold	8,700.00	8,550.00	11,280.00	19,890.00	
Total Gross Profit	2,085.28	2,308.29	2,485.77	4,921.82	

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e. Physicochemical Analysis of Dried Fish

The physicochemical analysis of dried fish processed using the dryer and the sun-drying methods has been studied through the Regional Standards and Testing Laboratory (RSTL) of DOST Region VI.

Tests such as Water Activity, NaCl as Salt, Microbial Analysis, and Proximate Analysis have been analyzed.

Tables 6 to 7 show that the analysis report showed that water is within the standard requirements. On the other hand, using the traditional brining method of Anchovies and the DOST-ITDI recommended brining method for Sardines did not pass the standard requirement for NaCl content.

Table 6. NaCl as Salt Value based on Analysis Report.

Type of Dried Fish	Brining Method	Drying Method		Standard Value
		Mechanical Drying	Sun-drying	
Sardine "Tabagak" (<i>Sardinella Spp</i>)	Traditional Brining 25% brine concentration, soaked for 2-3 hrs*	12.98	15.44	minimum of 12
		7.03		
Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Traditional Brining 25% brine concentration, soaked for 2-3 hrs*	6.16	7.12	minimum of 12
		23.26		
Split Red Mullet "Daing Salmonete" (<i>Mullus surmuletus</i>)	Traditional Brining 20% brine concentration, soaked for 1 hrs*	15.93	13.42	minimum of 12
		14.32		

**DOST-ITDI Recommended Brining Method*

Table 7. Water Activity based on Analysis Report.

Type of Dried Fish	Brining Method	Drying Method		Standard Value
		Mechanical Drying	Sun-drying	
Sardine "Tabagak" (<i>Sardinella Spp</i>)	Traditional Brining	0.7	0.71	maximum of 0.78
	25% brine concentration, soaked for 2-3 hrs*	0.74		
Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Traditional Brining	0.65	0.65	maximum of 0.78
	25% brine concentration, soaked for 2-3 hrs*	0.64		

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Split Red Mullet "Daing Salmonete" (<i>Mullus surmuletus</i>)	Traditional Brining	0.7	0.76	maximum of 0.78
20% brine concentration, soaked for 1 hrs*		0.71		
<i>*DOST-ITDI Recommended Brining Method</i>				

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

The multi-layer fish dryer, with a biomass furnace, has been designed to optimize the drying process of fish, targeting various species with different moisture content requirements. The technology incorporates features such as adjustable air temperature controls, displays for drying air temperature and relative humidity, and a drying chamber and trays made of food-grade materials. Results of the study show that the developed technology provides a more efficient, uniform drying process without the need for tray rotation and by accommodating different fish types and drying conditions. Using rice hulls as a biomass fuel source utilizes a waste product and ensures an eco-friendly and cost-effective operation. Its drying chamber can accommodate up to more than 100 kilograms of fish per batch.

The technology exceeds the 50% Heating System Efficiency requirement per PNS/BAFS 342:2022, demonstrating higher efficiencies based on in-house and AMTEC testing. It also adheres to environmental regulations and ensuring that the drying process does not expose the fish to contaminants, aligning with good manufacturing practices (GMP).

Results of the study show that the developed technology provides a more uniform drying process in terms of dried fish's air temperature and moisture content without the need for tray rotation, hence improving drying time and efficiency.

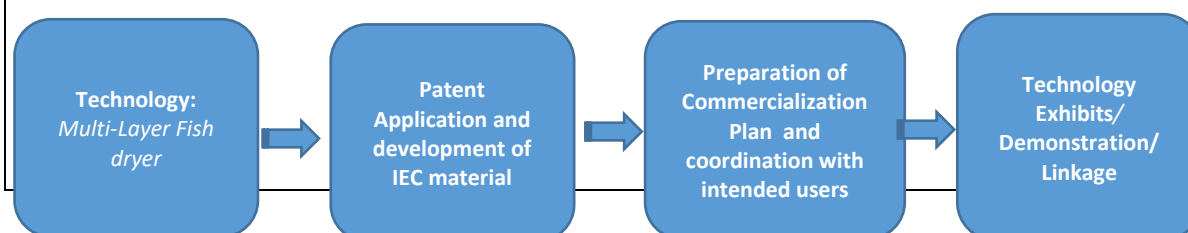
Considering the drying operating cost and dried fish selling price, the fish processor may gain up to Php4,000.00 gross profit in one drying batch.

Introducing the enhanced biomass-fired fish dryer to dried fish processors and the government of Capiz could revolutionize the local dried fish industry by providing a reliable and efficient drying solution. The dryer may be further developed by increasing its capacity and automating some of the dryer's functionalities.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

- 1 Patent Application
- 1 Product (Multi-layer fish dryer with a Biomass furnace)
- 2 People Services (Conduct training on the use of the dryer among intended users)
- 2 Partnerships (1 MOA with dried fish processor cooperative and a machine fabricator)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)



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(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

N/A

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

Gratitude is extended to the DOST-TAPI, for funding the study, and to CapSU the facilities and other resources.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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